MATH 211 Analysis III — Problem Session 6

Problem 1

For each of the following, prove that f is differentiable on its domain and compute Df.

a) $f(x,y) = (\sin x,\ xy,\ \cos y)$

b) $f(s,t,u,v) = (st+u^2,\ uv-s^2)$

c) $f(t) = \left(\log t,\ \frac{1}{1+t}\right)$

d) $f(r,\theta) = (r\cos\theta,\ r\sin\theta)$

Solution. All partials exist and continuous (on its domain). $\Rightarrow$ Everything is differentiable.

$$Df(x,y) = \begin{bmatrix} \cos x & 0 \\ y & x \\ 0 & -\sin y \end{bmatrix}, \qquad Df(s,t,u,v) = \begin{bmatrix} t & s & 2u & 0 \\ -2s & 0 & v & u \end{bmatrix},$$

$$Df(t) = \begin{bmatrix} \frac{1}{t} \\ \frac{-1}{(1+t)^2} \end{bmatrix}, \qquad Df(r,\theta) = \begin{bmatrix} \sin\theta & -r\sin\theta \\ r\cos\theta & r\cos\theta \end{bmatrix}.$$

Problem 2

Prove that $f(x,y) = \sqrt{xy}$ is not differentiable at $(0,0)$.

Solution.

$$f_x = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = 0,$$
$$f_y = \lim_{h\to 0} \frac{f(0,h)-f(0,0)}{h} = 0.$$

If f is differentiable then $Df = \begin{bmatrix} 0 & 0 \end{bmatrix}$. This implies

$$\lim_{(x,y)\to(0,0)} \frac{f(x,y)-f(0,0)+(0,0)}{\|(x,y)\|} = \lim_{(x,y)\to(0,0)} \frac{\sqrt{xy}}{\sqrt{x^2+y^2}} = 0.$$

But then for $x = y$,

$$\lim_{\substack{(x,x)\\ \to(0,0)}} \frac{\sqrt{|xy|}}{\sqrt{x^2+x^2}} = \lim_{(x,x)\to(0,0)} \frac{x}{2x} = \frac{1}{2}.$$

Problem 3

Prove the following function is not differentiable at $(0,0)$:

$$f(x,y) = \begin{cases} \dfrac{x^2+y^2}{\sin\sqrt{x^2+y^2}} & 0 < \|(x,y)\| < \pi \\ 0 & (x,y) = (0,0). \end{cases}$$

Solution.

$$\begin{aligned} f_x(0,0) &= \lim_{h\to 0} \frac{f(0+h,0)-f(0,0)}{h} \\ &= \lim_{h\to 0} \frac{h^2}{\sin|h|\cdot h} = \lim_{h\to 0} \frac{h}{\sin|h|} \qquad \text{does not exist} \end{aligned}$$

$\Rightarrow f(x,y)$ is not differentiable at $(0,0)$.

Problem 4

Let $r>0$, $f:\mathrm{Br}(0,r)\to\mathbb{R}$, and suppose that there exist an $\alpha>1$ such that $|f(x)|\le\|x\|^\alpha$ for all $x\in\mathrm{Br}(0,r)$.

a) Prove that f is differentiable at 0.

b) What happens to this result when $\alpha=1$?

Solution. For $x=0$, $|f(0)|\le 0$, $\Rightarrow f(0)=0$.

$$\lim_{h\to 0}\frac{f(he_k)-f(0)}{h} = \lim_{h\to 0}\frac{|f(he_k)|}{h} \le \frac{\|he_k\|^\alpha}{|h|} \le \frac{|h|^\alpha}{|h|} \xrightarrow[h\to 0]{} 0.$$

Then,

$$\frac{|f(x)-f(0)-Bx|}{\|x\|} = \frac{|f(x)|}{\|x\|} \le \frac{\|x\|^\alpha}{\|x\|} \to 0.$$

Differentiable at 0 with $Df=\begin{bmatrix}0 & \cdots & 0\end{bmatrix}$.

Problem 5

Prove that

$$f(x,y)=\begin{cases}\dfrac{x^4+y^4}{(x^2+y^2)^\alpha} & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0).\end{cases}$$

is differentiable on $\mathbb{R}^2$ for all $\alpha<\frac{3}{2}$.

Solution. For $\mathbb{R}^2\setminus\{(0,0)\}$, differentiable because partial derivatives exist and continuous.

For origin: $(x^4+y^4)\le(x^2+y^2)^2$.

$$|f(x,y)|\le\frac{(x^2+y^2)^2}{(x^2+y^2)^\alpha}=\|(x,y)\|^{4-2\alpha}.$$

For $\alpha<\frac{3}{2}$, we have $4-2\alpha>1$. By previous problem f is differentiable at $(0,0)$ with $Df(x,y)=\begin{bmatrix}0 & 0\end{bmatrix}$. Hence f is differentiable at $(0,0)$.

Problem 6

Prove that if $\alpha>\frac{1}{2}$, then

$$f(x,y)=\begin{cases}|xy|^\alpha\log(x^2+y^2) & (x,y)\neq(0,0)\\ 0 & (x,y)=(0,0).\end{cases}$$

is differentiable on $\mathbb{R}^2$.

Solution.

$$f_x(0,0) = \lim_{h\to 0} \frac{f(h,0)-f(0,0)}{h} = \lim_{h\to 0} |h|^\alpha \cdot \log(h^2) = 0,$$
$$f_y(0,0) = \lim_{h\to 0} \frac{f(0,h)-f(0,0)}{h} = \lim_{h\to 0} |h|^\alpha \cdot \log(h^2) \to 0.$$

Then if f is differentiable, $Df(x,y) = \begin{bmatrix} 0 & 0 \end{bmatrix}$. We calculate

$$\lim_{(x,y)\to(0,0)} \frac{f(x,y)-f(0,0)-D0}{\|(x,y)\|} = \lim_{(x,y)\to(0,0)} \frac{f(x,y)}{\|(x,y)\|}.$$

Let $r = \sqrt{x^2+y^2}$.

$$|xy| \le \frac{1}{2}(x^2+y^2) = \frac{1}{2}r^2.$$

$$\begin{aligned} \Rightarrow\ |f(x,y)| &\le \left(\frac{1}{2}r^2\right)^\alpha \log(r^2) \\ &= \left(\frac{1}{2}\right)^\alpha \cdot r^{2\alpha} \cdot \log(r^2). \end{aligned}$$

As $r \to 0$,

$$\frac{|f(x,y)|}{r} \le \left(\frac{1}{2}\right)^\alpha r^{2\alpha-1} \cdot \log(r^2) \to 0 \quad \text{as } r \to 0.$$

Problem 7

Prove that

$$f(x,y) = \begin{cases} \dfrac{x^3-xy^2}{x^2+y^2} & (x,y) \neq (0,0) \\ 0 & (x,y) = (0,0) \end{cases}$$

is continuous on $\mathbb{R}^2$, has first-order partial derivatives everywhere on $\mathbb{R}^2$, but f is not differentiable on $\mathbb{R}^2$.

Solution.

$$\begin{aligned} \lim_{(x,y)\to(0,0)} \frac{x^3-xy^2}{x^2+y^2} &= \lim_{(x,y)\to(0,0)} \frac{x(x^2-y^2)}{x^2+y^2} \\ &= \frac{|x|\,|x^2-y^2|}{|x^2+y^2|} \le |x| \to 0, \end{aligned}$$

thus f is continuous.

$$f_x(0,0) = 1, \qquad f_y(0,0) = 0.$$

If f is differentiable then $Df(0,0) = \begin{bmatrix} 1 & 0 \end{bmatrix}$.

$$\lim_{(x,y)\to(0,0)} \frac{f(x,y)-f(0,0)-D(1,0)}{\|(x,y)\|} = 0$$

$$\lim_{(x,y)\to(0,0)} \frac{x^3-xy^2-x(x^2+y^2)}{(x^2+y^2)^{3/2}} = 0$$

Simplifying the numerator,

$$\lim_{(x,y)\to(0,0)} \frac{-2xy^2}{(x^2+y^2)^{3/2}} = 0.$$

But for $x = y > 0$,

$$\lim_{(x,x)\to(0,0)} \frac{-2x \cdot x^2}{(2x^2)^{3/2}} = \lim_{x\to 0^+} \frac{-2x^3}{2\sqrt{2}\, x^3} = -\frac{1}{\sqrt{2}} \neq 0,$$

a contradiction. Hence f is not differentiable at $(0,0)$.

Problem 8

Let $T \in \mathcal{L}(\mathbb{R}^n, \mathbb{R}^m)$. Prove that T is differentiable everywhere on $\mathbb{R}^n$ with $DT(a) = T$ for all $a \in \mathbb{R}^n$.

Solution. Let T be represented by the matrix A.

$$\lim_{h\to 0} \frac{T(a+h) - T(a) - Ah}{\|h\|} = \lim_{h\to 0} \frac{T(a) + T(h) - T(a) - T(h)}{\|h\|} = 0.$$

$\Rightarrow T$ is differentiable everywhere in $\mathbb{R}^n$.